PRAKTIKUM BASIS DATA

MODUL 08

Scripting SQL (Create index, Create view, Analyze Table)

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL

Kerjakan studi kasus pada modul 8 (halaman 87). Sertakan rekaman layar untuk setiap nomor.

8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
   c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

JAWABAN

1. Desain database untuk suatu perusahaan dengan relasi antar tabel (Department, Employee, Vendor, Product Item)

```sql
CREATE TABLE department (
    department_id NUMBER PRIMARY KEY,
    department_name VARCHAR2(100),
    manager_id NUMBER,
    location_id NUMBER
);

CREATE TABLE employee (
    employee_id NUMBER PRIMARY KEY,
    last_name VARCHAR2(100) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER(10,2),
    commission_pct NUMBER(4,2),
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_emp_dept FOREIGN KEY (department_id) REFERENCES department(department_id)
);

CREATE TABLE vendor (
    vendor_id NUMBER PRIMARY KEY,
    vendor_name VARCHAR2(100) NOT NULL,
    phone_number VARCHAR2(20),
    address VARCHAR2(255)
);

CREATE TABLE product_item (
    item_id NUMBER PRIMARY KEY,
    item_name VARCHAR2(100) NOT NULL,
    price NUMBER(15,2),
    stock_quantity NUMBER,
    vendor_id NUMBER,
    CONSTRAINT fk_item_vendor FOREIGN KEY (vendor_id) REFERENCES vendor(vendor_id)
```

2. Analisis table-table dengan compute statistics (Sebelum Data Diisi)

```sql
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE vendor COMPUTE STATISTICS;
ANALYZE TABLE product_item COMPUTE STATISTICS;

INSERT INTO department VALUES (10, 'Administration', 101, 1500);
```

Script Output

Task completed in 0.152 seconds

```
Table PRODUCT_ITEM created.

Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.

Table VENDOR analyzed.

Table PRODUCT_ITEM analyzed.
```

3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

```sql
INSERT INTO department VALUES (10, 'Administration', 101, 1500);
INSERT INTO department VALUES (20, 'Marketing', 102, 1600);
INSERT INTO department VALUES (30, 'Purchasing', 103, 1700);
INSERT INTO department VALUES (40, 'Human Resources', 104, 1800);
INSERT INTO department VALUES (50, 'IT Support', 105, 1900);
INSERT INTO department VALUES (60, 'Public Relations', 106, 2000);
INSERT INTO department VALUES (70, 'Sales', 107, 2100);
INSERT INTO department VALUES (80, 'Software Engineering', 108, 2200);
INSERT INTO department VALUES (90, 'Executive', 109, 2300);
INSERT INTO department VALUES (100, 'Finance', 110, 2400);
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 10 in 0.066 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	101	1500
2	20	Marketing	102	1600
3	30	Purchasing	103	1700
4	40	Human Resources	104	1800
5	50	IT Support	105	1900
6	60	Public Relations	106	2000
7	70	Sales	107	2100
8	80	Software Engineering	108	2200
9	90	Executive	109	2300
10	100	Finance	110	2400

```sql
INSERT INTO employee VALUES (1, 'Warisman', 'warisman@company.com', 8000, 0.1, TO_DATE('2024-01-15', 'YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (2, 'Santoso', 'santoso@company.com', 7500, NULL, TO_DATE('2024-02-20', 'YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (3, 'Pratama', 'pratama@company.com', 9000, 0.15, TO_DATE('2023-11-10', 'YYYY-MM-DD'), 80);
INSERT INTO employee VALUES (4, 'Wijaya', 'wijaya@company.com', 6000, NULL, TO_DATE('2025-01-05', 'YYYY-MM-DD'), 50);
INSERT INTO employee VALUES (5, 'Kusuma', 'kusuma@company.com', 5500, NULL, TO_DATE('2025-03-12', 'YYYY-MM-DD'), 50);
INSERT INTO employee VALUES (6, 'Siregar', 'siregar@company.com', 12000, 0.2, TO_DATE('2022-08-22', 'YYYY-MM-DD'), 90);
INSERT INTO employee VALUES (7, 'Setiawan', 'setiawan@company.com', 6500, NULL, TO_DATE('2024-07-30', 'YYYY-MM-DD'), 10);
INSERT INTO employee VALUES (8, 'Nugroho', 'nugroho@company.com', 7200, 0.05, TO_DATE('2024-09-18', 'YYYY-MM-DD'), 70);
INSERT INTO employee VALUES (9, 'Saputra', 'saputra@company.com', 7100, 0.05, TO_DATE('2024-10-25', 'YYYY-MM-DD'), 70);
INSERT INTO employee VALUES (10, 'Hidayat', 'hidayat@company.com', 8500, NULL, TO_DATE('2023-05-14', 'YYYY-MM-DD'), 100);
```

pt Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 10 in 0.025 seconds

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	1	Warisman	warisman@company.com	8000	0,1	15-01-2024	80
2	2	Santoso	santoso@company.com	7500	(null)	20-02-2024	80
3	3	Pratama	pratama@company.com	9000	0,15	10-11-2023	80
4	4	Wijaya	wijaya@company.com	6000	(null)	05-01-2025	50
5	5	Kusuma	kusuma@company.com	5500	(null)	12-03-2025	50
6	6	Siregar	siregar@company.com	12000	0,2	22-08-2022	90
7	7	Setiawan	setiawan@company.com	6500	(null)	30-07-2024	10
8	8	Nugroho	nugroho@company.com	7200	0,05	18-09-2024	70
9	9	Saputra	saputra@company.com	7100	0,05	25-10-2024	70
0	10	Hidayat	hidayat@company.com	8500	(null)	14-05-2023	100

```sql
INSERT INTO vendor VALUES (101, 'PT Tekno Maju', '081234567890', 'Jakarta');
INSERT INTO vendor VALUES (102, 'CV Abadi Jaya', '081298765432', 'Bandung');
INSERT INTO vendor VALUES (103, 'Global Parts ID', '081345678901', 'Surabaya');
INSERT INTO vendor VALUES (104, 'Sentosa Elektronik', '081456789012', 'Semarang');
INSERT INTO vendor VALUES (105, 'Bintang Hardware', '081567890123', 'Yogyakarta');
INSERT INTO vendor VALUES (106, 'Mitra Sejahtera', '081678901234', 'Malang');
INSERT INTO vendor VALUES (107, 'Karya Mandiri', '081789012345', 'Solo');
INSERT INTO vendor VALUES (108, 'Sinar Mas Supplies', '081890123456', 'Denpasar');
INSERT INTO vendor VALUES (109, 'Data Lintas Nusantara', '081901234567', 'Medan');
INSERT INTO vendor VALUES (110, 'Mekar Tech', '082012345678', 'Makassar');

INSERT INTO product_item VALUES (1001, 'Laptop ThinkPad', 15000000, 50, 101);
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 10 in 0.014 seconds

	VENDOR_ID	VENDOR_NAME	PHONE_NUMBER	ADDRESS
1	101	PT Tekno Maju	081234567890	Jakarta
2	102	CV Abadi Jaya	081298765432	Bandung
3	103	Global Parts ID	081345678901	Surabaya
4	104	Sentosa Elektronik	081456789012	Semarang
5	105	Bintang Hardware	081567890123	Yogyakarta
6	106	Mitra Sejahtera	081678901234	Malang
7	107	Karya Mandiri	081789012345	Solo
8	108	Sinar Mas Supplies	081890123456	Denpasar
9	109	Data Lintas Nusantara	081901234567	Medan
10	110	Mekar Tech	082012345678	Makassar

```sql
INSERT INTO product_item VALUES (1001, 'Laptop ThinkPad', 15000000, 50, 101);
INSERT INTO product_item VALUES (1002, 'Mechanical Keyboard', 800000, 100, 102);
INSERT INTO product_item VALUES (1003, 'Wireless Mouse', 350000, 150, 102);
INSERT INTO product_item VALUES (1004, 'Monitor 24 Inch', 2500000, 40, 104);
INSERT INTO product_item VALUES (1005, 'Server Rack 42U', 12000000, 10, 109);
INSERT INTO product_item VALUES (1006, 'Network Switch 24 Port', 4500000, 25, 109);
INSERT INTO product_item VALUES (1007, 'Ergonomic Chair', 1200000, 80, 106);
INSERT INTO product_item VALUES (1008, 'Office Desk', 1800000, 60, 106);
INSERT INTO product_item VALUES (1009, 'UPS 1000VA', 1100000, 30, 105);
INSERT INTO product_item VALUES (1010, 'SSD 1TB NVMe', 1500000, 120, 103);
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

SQL | All Rows Fetched: 10 in 0.02 seconds

	ITEM_ID	ITEM_NAME	PRICE	STOCK_QUANTITY	VENDOR_ID
1	1001	Laptop ThinkPad	15000000	50	101
2	1002	Mechanical Keyboard	800000	100	102
3	1003	Wireless Mouse	350000	150	102
4	1004	Monitor 24 Inch	2500000	40	104
5	1005	Server Rack 42U	12000000	10	109
6	1006	Network Switch 24 Port	4500000	25	109
7	1007	Ergonomic Chair	1200000	80	106
8	1008	Office Desk	1800000	60	106
9	1009	UPS 1000VA	1100000	30	105
10	1010	SSD 1TB NVMe	1500000	120	103

4. Analisis table-table dengan compute statistics (Setelah Data Diisi)

```
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE vendor COMPUTE STATISTICS;
ANALYZE TABLE product_item COMPUTE STATISTICS;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3

Task completed in 0.147 seconds

```
Table DEPARTMENT analyzed.

Table EMPLOYEE analyzed.

Table VENDOR analyzed.

Table PRODUCT_ITEM analyzed.
```

5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.

```
CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;

PROMPT === ISI VIEW EMPVU80 ===
SELECT * FROM empvu80;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4

SQL | All Rows Fetched: 3 in 0.009 seconds

	ID_NUMBER	NAME	SALARY	DEPARTMENT_ID
1	1	Warisman	8000	80
2	2	Santoso	7500	80
3	3	Pratama	9000	80

6. Kemudian tampilkan struktur dari view empvu80.

```
DESC empvu80;
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4

Task completed in 0.422 seconds

```
View EMPVU80 created.

Name          Null?    Type
------------- -------- -------------
ID_NUMBER     NOT NULL NUMBER
NAME          NOT NULL VARCHAR2(100)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER
```

7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE. Kolom foreign key pada tabel employee adalah department_id.

```
CREATE INDEX idx_emp_dept ON employee(department_id);
```

Script Output | Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4

Task completed in 0.062 seconds

```
------------- -------- -------------
ID_NUMBER     NOT NULL NUMBER
NAME          NOT NULL VARCHAR2(100)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER

Index IDX_EMP_DEPT created.
```